PL4246 Final Report Instructions

Submission contents

1. Cover page: Contains title of project, your name,
   link to data repository, YouTube link for presentation
   (1 page max).
   1. *Your URLs should be "clickable"*
      , i.e., hyperlinked to the corresponding web page.
2. Report: The actual report is
   8-10 double-spaced pages
   (excluding abstract + reference list). You must include an abstract on the second page which is a brief summary of your project. No appendix/supplementary materials allowed (may be deposited in the repository), but you are allowed 2 additional pages after the main text to include any tables, figures where relevant (optional). The report ends with the reference list (no page limit).
   1. Abstract (100-150 words on its own page)
   2. Main report (min 8 pages, max 10 pages)
   3. Tables/figures (max 2 page, optional)
   4. References (no page limit)

Submit as **a single PDF** to Canvas. No other file type is permitted. Deadline: **13th November 2026, Friday, 11.59pm.**

It is your responsibility to ensure that the report is submitted on time. Please double check everything and do not leave it to the last minute. Late submissions receive a deduction in accordance to the syllabus, see "Policy regarding late work".

Style details

- font size: 12
- margins: 1 inch all around
- paper size: A4
- font type: a reasonable serif font such as Times New Roman or Cambria
- double spacing

Note: Submissions that do not adhere to the style guidelines will be immediately down-graded by half a grade.

Research Report (35%)

Description: This is a **8-10 double-spaced page** report that introduces and contextualizes the project within the existing psychological literature (introduction), explains the goals and objectives of the project and justifies the use of specific network measures (methods), and demonstrates the student's understanding of the broader implications of the results for psychology (results), and their reflection of the strengths and limitations of the project (discussion). The purpose of this document is to assess the clarity of your understanding of network concepts and your ability to apply them to answer new or interesting questions or puzzles within a psychological domain.

Tips

- remember that it is not about listing ALL of the analyses you have conducted - it is better to focus on the ones that really matter for your research question
- no project is perfect - please reflect on both its contributions and limitations (e.g., data quality, network construction challenges, limits of network measures, etc.)
- no appendix allowed in order to equalize the amount of space each student has to convey their results and interpretation
- it is OK to deviate from your preliminary report; but also do not simply plagiarize the contents
- if you use GenAI in any aspect of completing this report (writing, editing, programming, etc.), you must acknowledge its contributions. Refer to this article (https://libguides.nus.edu.sg/c.php?g=916984&p=7089387#s-lib-ctab-22880747-2) to see how AI use should be declared in academic work.

Grading rubric for Research Report

Features of a strong INTRODUCTION section

- Importance of research question is demonstrated; concise history/background provided; relevant literature is reviewed with appropriate emphasis and depth and integration, with clear connection to present study.
- Theoretical implications and/or connection between past and present work. How will the present study build on previous work? What is its intended contribution (especially with the application of network analysis)?

Features of a strong METHOD section

- Clear description of the network: operationalization of nodes and edges, edge type, network construction is described clearly and with justification or explanation of decisions and assumptions made

- Information about the data used to construct the network(s). If new data were obtained, then information about the data source, data collection, inclusion/exclusion criteria, etc., and other relevant information and procedures need to be included. Even if you are using pre-existing data (i.e., datasets 1-4), you should still provide a description of how the data was collected by the previous researchers.
- Pertinent information about any data cleaning, pre-processing steps should be included
- Clear description of the network analysis approach/plan, explanation or definition of network measures used, etc.

Features of a strong RESULTS section

- Basic descriptive information of the network(s) are included (i.e., number of nodes and edges, relevant network measures that provide an overview of the overall network structure)
- Network analysis results are clearly stated and presented, and well organized
- Tables / Figures – APA style with clear labels, referred to and explained in text, but placed after the main text, and only 2 pages max. Figures are legible and meaningful, able to convey useful information.
- (If relevant) Statistical findings reported in APA style (test statistic, df, p, effect size, etc.)

Features of a strong DISCUSSION section

- Reminds reader of study hypotheses (links back to introduction) with clear statement of support / nonsupport for hypothesis
- **Critical evaluation and interpretation of network analysis results that is thoughtful and considered**
  (not superficial)
- Theoretical implications to psychology are highlighted
- Qualifications / Limitations / Acceptance of negative results
- Clear statement of report's contributions
- Future research and directions, if relevant

REFERENCES

- References cited in text are appended at the end of the document, following APA style
- References used are appropriate (peer-reviewed journal articles/books in non-predatory journals/publishers, not Internet articles)

PRESENTATION & WRITING STYLE

- Orderly presentation of information
- Abbreviations/formatting, statistics reporting (if any), etc., conform to APA style
- Active voice, unbiased tone
-

Economy of expression - direct declarative sentences with simple words, no excessively flowery language

- Readable for non-expert (in the research topic and in network analysis)

Grading System

- A report with a mark of A/A- and above would have met most if not all of the criteria listed above; with no major flaws/omissions.
- A report with a mark of B+ would have met most of the criteria listed above; but with at least 1 major flaw/omission.
- A report with a mark of B would have met some of the criteria listed above; but with quite a few major flaws/omissions.
- A report with a mark of B- and below would have met some of the criteria listed above; but with too many flaws/omissions that undermine the work substantially.

Final Comments

1. **Do not view the above as a checklist**
   . Which features you focus on in your report depends on a number of factors: your specific study design, network analysis choices, data type, etc. Excellent reports present specific pieces of information in a way that is deeply contextualized and integrated given the specific characteristics of the project. They highlight the most important components of the study without compromising on comprehensibility. Weak papers usually try to include every single detail and as a result appear superficial, disorganized and unfocused.
2. From previous experience, the best papers are those with a
   clear research question or problem statement
   that can be clearly addressed with the tools of network science. This requires both a deep understanding of the topic and of network science concepts, and developing the ability to translate or make connections between network science metrics and theoretical ideas.
3. **Write your own paper.**
   While NUS does not explicitly forbid the use of technologies such as ChatGPT, it is important to note that if you do make use of AI technologies to help you complete the assignments for this module, it can be construed as plagiarism if you do not properly cite the AI tool and acknowledge its contributions to the specific parts of the assignment. Please see "Statement on the Use of AI Tools" and "Statement on Plagiarism" in the Syllabus for more information. If there is any suspicion of plagiarism, the reports will be further accessed by a plagiarism software to decide if a penalty will be imposed.
4. Try to
   think deeply about your project
   : What are the most important details necessary for someone to understand what the project is trying to accomplish? What was the most interesting and potentially impactful results obtained? What are the most substantial implications and/or flaws that need to be discussed in the report?

5. Any tables and figures should be placed after the main text so that they do not count toward the 10-page limit. They need to be clearly described with a meaningful caption and meaningfully referred to and discussed in the main text, not just put there for the sake of having a figure/table. They should not take up more than 2 pages.

Data Repository (5%)

Description: All data, analysis scripts, relevant materials and output (e.g., if you use SPSS/JASP for further analysis) should be clearly organized and prepared in a manner that would enable a future researcher to replicate the analysis conducted by the student.

Repositories

Recommended: Open Science Framework (OSF) - https://osf.io/, but OneDrive, Dropbox, or Google Drive is also fine.

If you are technically savvy, Github - https://github.com/

Examples

- spreadr - https://osf.io/a9bv6/
- inverse preferential attachment - https://osf.io/e8gwr/

Grading rubric for Data Repository

Organization and clarity

- are all relevant data, scripts, supplementary material in the repository?
- are your project files given clear, meaningful names/labels?
- do you provide a readme file or a description of your repository to explain its contents?
- are your R scripts (or Rmd/html) well commented so that it is clear what you have done?

Replicability and accuracy

The marker will download your repository as a zipped file and attempt to run your R script to see if the analysis can be repeated. Your R script will also be briefly reviewed for any errors or inaccuracies.

Tips

- All libraries used need to be declared at the beginning of your code so that the marker can ensure that these libraries are installed in their own R instance.
- In codes that read files in,
  you must ensure that the file paths are "generic" and made relative to the main R script (i.e., they will work even if a naive person downloads your repository into their computer and sets the working directory to the folder of your data and scripts)
- Either Rmd or R files are fine. A html rendering of the Rmd is also permitted.
- Useful project management and organization tips (highly recommended for "1. Organization and clarity"):
  https://slides.djnavarro.net/project-structure
- Try to put yourself in the shoes of someone who is unfamiliar with your project (but has the technical ability to understand it and wants to build on your research). How would you present your analysis in a way that they understand what you have done?
- Because it can be time consuming to recreate the network construction pipeline (especially if you are using raw data that is very large in size, or generating several different networks, or the pipeline is very complex), you may save the final network objects that you are using for your analysis for the examiner to load the networks easily. But your network construction process should still be clearly documented in your repository.

Presentation (10%)

Description: The format of the video can in any form (e.g., a voice-over with a PowerPoint presentation or Zoom recording is completely acceptable). The length of the video should be between 5 (min) to 10 (max) minutes. The goal of the video is to provide a clear, accessible overview of ONE network science concept that was applied in your research project: *what* is the network concept about, and *why* or *how* was it relevant to your research question(s) or topic.

Examples

- **NetSci 2025 Educational Video Competition**
  2nd place, "Spaghetti, Meatballs & Blackouts": A Tasty Lesson in Network Robustness, Luca Santagata
  https://youtu.be/cgbusiws-Y0

- **NetSci 2024 Educational Video Competition**
  3rd place, How to study the influence that we have on each other, that makes social behaviours propagate in our society?, Helcio Felipe, Timur Naushirvanov and Elsa Andres.
  https://www.youtube.com/watch?v=GrNiHQ134MA

Grading rubric for Presentation

Clarity, accuracy, and accessibility

- does your presentation:
  - describe the network science concept clearly, in a way that someone with little network science exposure can understand?
  - describe the network science concept accurately, without any errors?
  - clearly showcase how the network science concept is relevant to or appropriate for your research question or topic?

Creativity and innovation

- is the presentation interesting, engaging, and creative?
- this is in terms of the content and delivery, not the production quality/implementation

Tips

- you can keep your YouTube link un-searchable and then delete it completely after your grades are released if you are concerned about privacy
- my recommendation is to stick with a single network science concept, but you may increase it to two if you think it strengthens your presentation
- as you can see from the examples, it is not about the production quality of the video that matters, but the way in which the presenters conveyed the information to the audience
- a low tech video (i.e., voiceover + slides) is completely acceptable
- note that this is NOT a straight presentation of your research report! We are assessing your ability to explain network science ideas, in your very own words, to someone who does not know anything about networks.